

SOOHA SHIN

Computational Designer | Sydney, Australia

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SUMMARY

Computational Design graduate from UNSW (WAM 85) with research experience at Arup. I build parametric workflows, urban analysis tools, and robotic fabrication pipelines using Grasshopper, Rhino, Python, and full-stack web. My graduation project was featured on Dezeen and won the UNSW Computational Design Award 2025.

SKILLS

Parametric & BIM: Grasshopper, Rhino, Revit, AutoCAD, Kangaroo, Karamba3D

Programming: Python, Flask, HTML, CSS, JavaScript, Three.js, Mapbox, REST APIs

Data & AI: Machine learning, Computer vision, Geospatial analysis, Roboflow, YOLO, OpenAI API

Fabrication: UR5, KUKA KR70, Custom G-code, Laser cutting, 3D printing (FDM)

Collaboration: GitHub, Autodesk Construction Cloud (ACC), Figma, Adobe Creative Suite

EXPERIENCE

Research Collaborator, Arup (Supervisor: Haico Schepers)

2025

- Built a Python workflow combining SUEWS urban microclimate modelling with regression ML, generating site-specific hourly climate datasets for building energy simulation across Sydney LGAs.
- Cut prediction errors of up to $\pm 40\%$ in standard simulation workflows by training on surrounding weather stations and urban surface data, validated using Leave-One-Station-Out testing.
- Project featured in Dezeen's UNSW School Show 2025 coverage.

Student Peer Mentor, UNSW Built Environment

2025

- Weekly technical mentoring in Grasshopper, parametric modelling, and computational scripting for undergraduate Built Environment students.

SELECTED PROJECTS

Urban Lens | Flask, OpenAI, Three.js, REST APIs, GIS

- An AI-driven planning platform that evaluates uploaded STL building geometry against real LGA zoning datasets, returning compliance and site analysis across four assessment categories.

Never-Ending Lines (Robotic Carving) | Grasshopper, Kangaroo, UR5 Robotic Arm

- Custom Grasshopper toolpaths executed by a UR5 robotic arm across three ceramic prototypes, investigating computational fabrication and craft.

Material Quantification via Computer Vision | Roboflow, YOLOv11, Flask

- Instance segmentation model (98.2% mAP, 97.4% precision) classifying five building materials from facade imagery; aligned with UN SDGs 9 and 11 for sustainability reporting.

Urban Envelopes | Grasshopper, OpenStreetMap, Three.js

- Live web platform pulling real OpenStreetMap data into Grasshopper to generate parametric building envelopes, population density, and green space analysis on actual Sydney geography.

EDUCATION

Bachelor of Computational Design | University of New South Wales, Sydney | Conferred June 2026 | WAM: 85

RECOGNITION

- Computational Design Award 2025, UNSW Built Environment Graduation Exhibition
- Dezeen International, UNSW School Show 2025 (graduation project featured in leading architecture and design publication)

LANGUAGES

English (native) | Korean (native)

REFERENCE

Daniel Yu, Deputy Discipline Director, Computational Design, UNSW Sydney | daniel.yu@unsw.edu.au